

Properties of determinants.

Proposition 1 Let A be an $n \times n$ matrix and E be an elementary matrix
Prove that

$$\det(EA) = \det(E) \det(A)$$

Proof:

Case ① Suppose that E corresponds to the row operation
 $R_i \rightarrow R_i + \alpha R_j$.

Then, write

$$A = \begin{pmatrix} R_1 \\ \vdots \\ R_i \\ \vdots \\ R_j \\ \vdots \\ R_n \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} R_1 \\ \vdots \\ \alpha R_j \\ \vdots \\ R_j \\ \vdots \\ R_n \end{pmatrix}$$

and observe that

$$EA = \begin{pmatrix} R_1 \\ \vdots \\ R_i + \alpha R_j \\ \vdots \\ R_j \\ \vdots \\ R_n \end{pmatrix}$$

Thus by a result proved earlier,

$$\begin{aligned} \det(EA) &= \det(A) + \det(B) \\ &= \det(A) + \alpha \det \begin{pmatrix} R_1 \\ \vdots \\ R_j \\ \vdots \\ R_j \\ \vdots \\ R_n \end{pmatrix} \\ &= \det(A) + 0 \\ &= \boxed{\det(A)} \end{aligned}$$

Also, $\det(E) \det(A) = 1 \cdot \det(A) = \boxed{\det(A)}$

Thus, in this case, $\boxed{\det(EA) = \det(E) \det(A)}$

Case ② Suppose that E corresponds to the row operation

$$R_i \rightarrow \alpha R_i$$

Then,

$$E = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & & & 0 \\ 0 & & \ddots & & \\ 0 & & & \alpha & \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix}, \quad \det(E) = \alpha.$$

and that

$$EA = \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ \alpha R_i \\ \vdots \\ R_n \end{pmatrix}$$

$$\text{Thus } \det(EA) = \det \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ \alpha R_i \\ \vdots \\ R_n \end{pmatrix} = \alpha \det(A) = \det(E) \det(A)$$

Case ③ Let E be the elementary matrix corresponding to the row operation $R_i \leftrightarrow R_j$

Then, $\det(E) = -1$ and that $\det(EA) = -\det A$.

Thus $\det(EA) = -\det(A) = \det(E) \det(A)$.

This proves the Proposition.

Theorem Let A, B be two $n \times n$ matrices. Then

$$\det(AB) = \det(A) \det(B)$$

Proof

Case ① Let A be invertible. Then there are elementary matrices $E_1, E_2, \dots, E_n$ such that

$$E_1 E_2 \cdots E_n = A \quad (\text{why?})$$

$$\begin{aligned} \text{Thus } \det(A) &= \det(E_1 E_2 \cdots E_n) \\ &= \det(E_1) \det(E_2 \cdots E_n) \quad (\text{Use proposition!}) \\ &= \det(E_1) \det(E_2) \det(E_3 \cdots E_n) \\ &\quad \vdots \\ &= \det(E_1) \det(E_2) \cdots \det(E_n) \end{aligned} \quad \left(\begin{array}{l} \text{Use induction} \\ \text{and write a} \\ \text{rigorous proof} \end{array} \right)$$

Since $AB = E_1 E_2 \cdots E_n B$, we have

$$\begin{aligned} \det(AB) &= \det(E_1 E_2 E_3 \cdots E_n B) \\ &= \det(E_1) \det(E_2 E_3 \cdots E_n B) \\ &= \det(E_1) \det(E_2) \det(E_3 \cdots E_n B) \\ &\quad \vdots \\ &= \det(E_1) \det(E_2) \cdots \det(E_n) \det B \\ &= \det(A) \det(B). \end{aligned} \quad \left(\begin{array}{l} \text{Use induction} \\ \text{and write a} \\ \text{rigorous proof} \end{array} \right)$$

Thus, if A is invertible, we have proved that $\det(AB) = \det(A) \det(B)$

Case ② Suppose that A is not invertible.

Then, there are matrices $E_1, E_2, \dots, E_n$ such that

$$E_1 E_2 \cdots E_n A = J \quad \text{.....? Reduced row echelon form}$$

Now, $\det(E_1 E_2 \dots E_n A) = \det(E_1) \det(E_2) \dots \det(E_n) \det(A)$

and $\det(J) = 0$ (why?)

Since $\det(E_i) \neq 0$ for each i , we obtain $\det(A) = 0$

H.W. Write a proof and verify that $\det(A) \neq 0$ if and only if A is invertible.

Observe that $E_1 E_2 \dots E_n AB = JB$

$$\Rightarrow \det(E_1 \dots E_n AB) = \det(JB)$$

The last row of JB is a Zero row (why?). Therefore,
 $\det(JB) = 0$.

Thus, $0 = \det(JB) = \det(E_1 \dots E_n AB) = \det(E_1) \dots \det(E_n) \det(AB)$

Since $\det(E_i) \neq 0$ for each i , $\det(AB) = 0$.

Thus $0 = \det(A) = \det(AB)$ and consequently, we obtain
 $\det(AB) = \det(A) \det(B)$.

This Proves the theorem



Transpose of a matrix

Let $A = (a_{ij})$ be any $m \times n$ matrix. The transpose of A , denoted by A^T , is defined as the $n \times m$ matrix

$$A^T = (x_{ij}), \text{ where } x_{ij} = a_{ji}$$

Examples: @ $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \longrightarrow (1\ 2\ 3)$

$$\textcircled{b} \begin{pmatrix} 1 & 4 & 7 \\ 3 & 2 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 3 \\ 4 & 2 \\ 7 & 4 \end{pmatrix}$$

$$\textcircled{c} \begin{pmatrix} a & b & c \\ d & e & f \\ h & i & j \end{pmatrix} \longrightarrow \begin{pmatrix} a & d & h \\ b & e & i \\ c & f & j \end{pmatrix}$$

Rows of A become columns of A^T and columns of A become rows of A^T

Diagonal entries of A and A^T are the same!

Easy to prove facts. A be an $m \times n$ matrix, B be an $n \times p$ matrix.

$$\textcircled{1} (A^T)^T = A : A = (a_{ij}), A^T = (a_{ji})_{ij}, (A^T)^T = (a_{ij})_{ij}.$$

$$\textcircled{2} (AB)^T = B^T A^T.$$

The ij^{th} entry of AB is $\sum_{j=1}^n a_{ij} b_{jk}$ and ji^{th} entry of AB is $\sum_{l=1}^n a_{jl} b_{li}$

Thus ij^{th} entry of $(AB)^T$ is $\sum_{l=1}^n a_{jl} b_{li}$ and ij^{th} entry of $B^T A^T$ is

$$\sum_{l=1}^n a_{jl} b_{li}$$

Are they the same?

Where, $B^T = (y_{ij}), A^T = (x_{ij})$

$$y_{ij} = b_{ji}, x_{ij} = a_{ji}$$

$$\sum_{k=1}^n y_{ik} x_{kj} =$$

$$\sum_{k=1}^n b_{ki} a_{jk}$$

$$a_{j1} b_{1i} + a_{j2} b_{2i} + \dots + a_{jn} b_{ni} = b_{1i} a_{j1} + b_{2i} a_{j2} + \dots + b_{ni} a_{jn}$$

③ Let E be an elementary matrix ($n \times n$). Then
 $\det(E) = \det(E^T)$

Proof

① Let E be the matrix that performs the row operation

$$R_i \rightarrow R_i + \alpha R_j$$

Then $E = I_n + \alpha E_{ij}$. Therefore $E^T = (I_n + \alpha E_{ij})^T$
 and $\det(E) = 1$ (See earlier lectures) $= I_n^T + \alpha E_{ij}^T$
 $= I_n + \alpha E_{ji}$

So, E^T is the matrix that performs the row operation

$$R_j \rightarrow R_j + \alpha R_i$$

Thus $\det(E^T) = 1$

Hence $\det(E) = 1 = \det(E^T)$.

② If E is the matrix obtained by performing the row operation $R_i \rightarrow \alpha R_i$ then

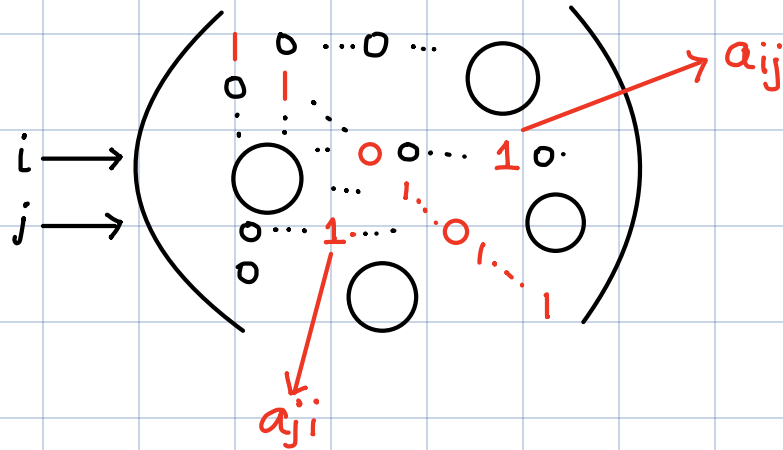
$$E = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \text{ and therefore } E = E^T.$$

Therefore $\det(E) = \det(E^T)$.

③ If E is the matrix obtained by switching rows $R_i \leftrightarrow R_j$ then

$$E = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$$

$$i < j$$



Then

$$E^T = I_n^T - E_{ii}^T - E_{jj}^T + E_{ij}^T + E_{ji}^T$$

$$\Rightarrow E^T = I_n - E_{ii} - E_{jj} + E_{ji} + E_{ij} = E.$$

Therefore $\det(E^T) = \det(E).$